$x_1 + 2x_3 = 0$

$\Rightarrow x_1 = -2x_3$

$x_1 + x_2 + x_3 = 0$

$\Rightarrow x_2 - x_3 = 0$

$\Rightarrow x_2 = x_3$

$$\therefore X = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -2x_3 \\ x_3 \\ x_3 \end{pmatrix} = \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix} x_3$$

Choose $x_3 = 1 \quad \therefore X = \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix}$

or

$$\begin{pmatrix} -1 & 0 & -2 \\ 1 & 1 & 1 \\ 1 & 0 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} -1 & 0 & -2 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$\Rightarrow -x_1 - 2x_2 = 0$

$x_2 - x_3 = 0$

Choose $x_3 = 1$

$\therefore x_1 = -2$

$x_2 = 1$

x_3

$$\therefore X = \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix}$$

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